

PHYS 515: Python for Data Science and Scientific Computing

Spring 2026 End-Term Project Datasets

University of Southern California

Project Dataset Overview

This document provides a list of datasets for the end-term projects.

#	Project Title	Dataset
1	Cancer Genomics: Multi-Class Classification from RNA-Seq Data	TCGA Pan-Cancer RNA-Seq
2	Exoplanet Detection: Time-Series Classification with Extreme Imbalance	Kepler Labeled Time Series
3	Earthquake Magnitude Prediction: Geospatial & Temporal Regression	USGS Global Seismic Events (2000–2025)
4	Spotify Hit Predictor: Popularity Regression & Recommendation Systems	Spotify 1 Million Tracks
5	S&P 500 Stock Market Analysis: Time-Series Forecasting & Crash Detection	S&P 500 Historical Data
6	CERN Particle Hunter: Supersymmetric Particle Detection	LHC MultiJet Primary Dataset
7	Climate Oracle: Temperature Anomaly Forecasting & Global Warming Analysis	Berkeley Earth Climate Data
8	Fraud Hunter: Real-Time Credit Card Fraud Detection with Extreme Imbalance	Credit Card Fraud Detection

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#	Project Title	Dataset
9	Heart Disease Prediction: Clinical Risk Assessment & Medical Decision Support	Heart Disease Dataset - 3000 Records
10	Loan Default Risk Modeling: Financial Risk Assessment & Credit Scoring	Loan Default Prediction Dataset

Dataset Access

All datasets are publicly available on Kaggle (<https://www.kaggle.com>). Students must create a free Kaggle account to download datasets. Each dataset link is provided in the table above.